- The warm temperature or the sub-tropical climate of the Northern zone gives it cold winter seasons and hot summer seasons. The Southern tropical climate zone is warmer than the North and does not have a clear cut winter season.
- The Southern zone has the midday Sun almost vertically overhead at least twice every year and the Northern zone does not have the mid-day Sun vertically overhead during any part of the year.
- There are various factors which influence the climate of India. Location and latitude plays an important role in affecting climate of India. Tropic of Cancer divides, India into tropical and sub-tropical climatic regions. Indian ocean influences the climate of peninsular.
- Himalayan range protects India from bitterly cold and dry winds from Central Asia and moreover acts as barrier in bringing monsoonal rainfall. Heating of interior part during summer attracts monsoon winds and cause monsoon rainfall. In winter Western disturbance cause snowfall in mountains and rainfall in plains.

Seasons in India

Indian climate is characterised by distinct seasonality. *Indian Meteorological Department (IMD) has recognised the following four distinct seasons*

i. The cold season or winter season.
ii. The hot weather season or summer season.
iii. The South-West monsoon season or rainy season.
iv. The season of the retreating monsoon or cool season.

NATURAL VEGETATION OF INDIA

India is a land of great variety of natural vegetation. Himalayas are marked with temperate vegetation; Western Ghats and Andaman and Nicobar islands have tropical rain forests; deltaic regions have tropical forests and mangroves; desert and semi desert areas are known for variety of bushes and thorny vegetation.

Indian vegetation can be divided into the following groups

Tropical Forests

Tropical forests are divided into–Moist Forest and Dry Forest

Moist Forest

Moist forest can be classified as :

Tropical Wet Evergreen Forests

- It is found in the areas where the annual rainfall exceeds 250 cm.
- The annual temperature is about 25°-27°C, the average annual humidity exceeds 77% and the dry season is distinctly short.
- It includes areas—the Western side of the Western ghats, a strip running from North-East to South-West direction across Arunachal Pradesh, upper Assam, Nagaland, Andaman and Nicobar Island mahogony, eboagle.
- Species of trees found in this forests are white cedar, mesua, jamun, hopea, mahogony, ebony etc.

Tropical Semi-Evergreen Forests

- These are found in the region where the annual rainfall is 200-250 cm.
- The mean annual temperature varies from 24°-27°C and the relative humidity is about 75%.
- It includes areas—Western coast, Assam lower slopes of the Eastern Himalayas, Odisha and Andamans.
- Species of trees—aini, semul, kadam, rosewood, kusum etc.

Tropical Moist Deciduous Forests

- These are found in the areas having rainfall of 100 to 200 cm per annum.
- Mean annual temperature of about 27°C the average relative humidity of 60 to 70%.
- It include areas—along the Western ghats surrounding the belt of evergreen forests, a strip along the Shiwalik range including Terai and Bhabar from 77°E to 88°E, hills of Eastern Madhya Pradesh, Chhattisgarh, Chhotanagpur and part of Odisha and West Bengal.
- Species of trees found in this forests are teak, sal, laurel, white chuglam, badam, mahua and bamboo etc.

Littoral and Swamp Forests

- These forests occur in and around the deltas, estuaries and creeks.
- Species of trees found are—*sundari, rhizophora, srwpines, sonnoratic* etc.
- These forests can survive and grow both in fresh as well as brackish water.

Dry Forest

Dry forest can be classified as :

Tropical Dry Evergreen Forests

- These are found along the coasts of Tamil Nadu, these forests occur in short stature.
- Annual rainfall is about 100 cm and the mean annual temperature is about 28°C.
- The mean humidity is about 15%, species of trees found here are khirni, jamun, tamarind, neem etc.

Tropical Dry Deciduous Forests

- These are similar to moist deciduous forests and shed their leaves in dry season.
- These forests can grow in areas of even less rainfall of 100-150 cm per annum.
- Species of trees— *teak, axlewood, tendu, palas, bel* etc.

Tropical Thorn Forests

- These forests generally occur in the area of low rainfall and high temperature.
- Species of trees found are—*Khair, Neem, Babul, Cacti, Palas* etc.

- The areas are North-Western parts of the country including Rajasthan, South-Western Punjab, Western Haryana, Kutch etc.

Sub-tropical Forest

Sub-tropical forest are of three types

Sub-tropical Broad-leaved Hill Forests

- These forests occur in the Eastern Himalayas to the East of 88°E longitude at altitudes varying from 1000 to 2000 m.
- The mean annual rainfall is 75 cm to 125 cm, average annual temperature is 18°-21°C. They form luxurious forests of evergreen species.
- Species of trees—Oaks, Chestnuts, Sals and Pines (on lower and higher margin respectively) etc.
- They also occur in the Nilgiri and Palni Hills at 1070-1525 m above sea level. These forests are generally called *shales.*

Sub-tropical Moist Pine Forests

- They are found at the height of 1000 to 2000 m above sea level in the Western Himalayas between 73°E and 88°E longitudes.
- **Chir** is the most dominant tree.

Sub-tropical Dry Evergreen Forests

- Found in the Bhabar, the Shivaliks and the Western Himalayas upto about 1000 m above sea-level. Rainfall is between 50 to 100 cm.
- Olive, Acacia, Modesta and Pistacia are the important species of trees.

Temperate Forest

Temperate forest is further divided into 3 types futher are of three types

Montane Wet Temperate Forests

- The forests grow at a height of 1800 to 3000 m above sea level. The mean annual rainfall is 150 cm to 300 cm, the mean annual temperature is about 11°C-14°C and the average relative humidity is over 80%.
- Species of trees—deodar, chilauni, Indian chestnut, birch, blue pine etc.
- They are found in the higher hills of Tamil Nadu and Kerala, in the Eastern Himalayan region to the East of 88° E longitude.

Himalayan Moist Temperate Forests

- These forests are mainly composed of coniferous species such as pines, cedars, silver, firs, spruce etc are most important trees.
- These forests occur in the temperate zone of the Himalayas between 1500 and 3300 m. Rainfall varies from 150 cm to 250 cm.

Himalayan Dry Temperate Forests

- These are coniferous forests with xerophytic shrubs. Deodar, chilgoza, oak, olive etc are the main trees.
- Such forests are found in the inner dry ranges of the Himalayas.

MANGROVES

Mangroves are very specialised forest ecosystem of tropical and sub-tropical regions of the world bordering sheltered sea-coasts. They occur all along the Indian coastline in the sheltered estuaries, tidal creeks, backwaters, salt marshes and mudflats.

Mangroves are dominated by salt tolerant halophytic plants of diverse structure and are invaluable **marine nurseries** for a large variety of fish and other marine fauna. They support a large variety of birds, amphibians and many other local arboreal, benthic and water creatures.

Mangroves have a dense network of aerial roots, which help to aerate the root system and anchor the tree. Sundari is widespread in sunderbans, screw pines, canes and palms are common in deltas, cracks are often lined with Nipa.

SOIL

- Soil is formed when rocks are broken down by the action of wind, water and climate. This process is called **weathering**.
- Soil forms different layers of particles of different sizes called Horizons. Each layer is different from the other in thickness texture, colour and chemical composition.

Major Soils of India

On the basis of genesis, colour, composition and location, the soils of India have been classified into the following types

Alluvial Soil

- They cover the largest area in India (40%) and is the most important soil from agricultural point of view. Alluvial soil is widespread in the Northern plains and the river valleys. Through a narrow corridor in Rajasthan, they extend into the plains of Gujarat.
- Geologically, the alluvium is divided into **new alluvium** which is known as **Khadar** and old alluvium, as Bhangar. The newer alluvium is sandy and light coloured, whereas, older alluvium is more clayey, dark coloured and contains lime concretions.
- The conglomerate deposits in piedmont area are generally known as Bhangar. This soil is suitable for rice, wheat, sugarcane, oil seeds and jute cultivation.

Black Soil/ Regur Soil

- The principal region of black soil is the Deccan plateau and its periphery. This is formed from Deccan basalt trap rocks and occur in areas under the monsoon climate, mostly of semi-arid and sub-humid types.
- This soil is characterised by dark grey to black colour, high swelling and shrinkage, plasticity, deep cracks during summer and poor status of organic matter, nitrogen and phosphorus while this is rich in lime, iron, magnesia and alumina.
- Impeded drainage and low permeability are the major problems. Cotton is mostly grown on this soil.

Red and Yellow Soils

- These soils are derived from granite, gneiss and other metamorphic rocks. These soils are formed under well drained condition. These soils are high textured and contain low soluble salts.
- They are slightly acidic to slightly alkaline, well drained with moderate permeability. They are also poor in nitrogen, phosphorus, lime, humus etc.
- Red soils cover a large part of Tamil Nadu, Karnataka, Andhra Pradesh, Telangana, Chhattisgarh, Jharkhand, Madhya Pradesh and Odisha.

Laterite Soil

- Laterite soil is peculiar to India and some of the tropical countries where there is high temperature and heavy rainfall with alternating wet and dry periods. During rainfall silica is leached downwards and iron and aluminum oxide remains in the top layers.
- In hilly areas they are more acidic than that of low areas. They are less fertile and poor in humus, nitrogen, phosphate and calcium.
- Lateritic soils are found in Odisha, West Bengal, in some parts of Andhra Pradesh, Tamil Nadu, Kerala, Jharkhand and Maharashtra.

Desert Soil

- In the North-Western part of India, desert soil occurs over the major parts of Rajasthan, South of Haryana and Punjab and Northern part of Gujarat. The soil in the plains are mostly derived from alluvium and pale brown to brown to yellow brown and fine sandy to loamy fine sand and is structureless.
- The clay contents low and presence of alkaline Earth carbonates is an important feature. By increasing the water holding capacity, the productivity of the soil can be increased, which involves addition of organic matter and clay.

Swampy/Peaty Soil

Peaty soil originate in areas of heavy rainfall, but inadequate drainage facility. This soil is usually found at the foot hills and extend in strips of varying widths at the foot of Himalayas in Jammu and Kashmir, Uttarakhand, Uttar Pradesh, Bihar and West Bengal.

Saline Soil

Saline soil is formed due to accumulation of soluble salts which consists of chlorides and sulphates of calcium and magnesium. In India, areas around 7 million hectares are salt affected distributed in different states.

Forest Soil

As the name suggests, forest soil is formed in the forest areas, where sufficient rainfall is available. The soil vary in structure and texture depending on the mountain environment where these are formed.

Soil Erosion and Degradation

- The destruction of the soil cover is described as soil erosion, while decrease in its fertility is **soil degradation**. Wind and water are powerful agents of soil erosion because of their ability to remove soil and transport it.
- **Wind erosion** is significant in arid and semi-arid regions. In regions with heavy rainfall and steep slopes, erosion by running water is more significant.

AGRICULTURE IN INDIA

India is a vast country endowed with a great variety of natural environments and thus provides conditions for a large number of crops to be grown in various parts.

Types of Farming

Various geographical, physical and socio-economic factors are responsible for giving birth to different types of farming in different parts of the country.

Subsistence Farming

Farmers cultivate small and scattered holdings with the help of draught animal and family members. The tools and techniques used are primitive and simple and main focus is on food crops. The farmers and his family members consume the entire farm production.

Plantation Farming

It involves growing and processing of a single cash crop purely meant for sale. It is capital intensive and the other necessary things needed are vast estate, managerial ability, technical know how, fertilizer, good transport facilities, processing factory etc. This type of agriculture is mainly practiced in Assam, sub-Himalayan West Bengal and in Nilgiri, Anaimalai and Cardamom hills in South.

Shifting Agriculture

- It is practised by the tribals in the forest areas of Assam, Meghalaya, Nagaland, Manipur, Tripura, Mizoram, Arunachal Pradesh, Odisha, Madhya Pradesh and Andhra Pradesh. In this type of agriculture, a piece of forest land is cleared mainly by tribal people with felling and burning of trees and crops are grown.
- Dry paddy, buck wheat, maize, small millets, tobacco and sugarcane are the main crops grown under this type of agriculture. This is a very primitive method of cultivation which results in large scale deforestation and soil erosion especially on the foothill sides.

Organic Farming

- A new trend of farming in which all input used for farming are natural, no chemicals are used.
- Green manures and compost are used. Sikkim is the first organic state of India.

Cropping Seasons

Three types of cropping seasons are found in India

Kharif	It requires much water, long hot weather for their growth, grown in June with the arrival of South-East monsoon. e.g. rice, jowar, maize, cotton, groundnut, jute, tobacco, bajra, sugarcane, pulses etc.
Rabi	Grown in winter, required cool climate during growth and warm climate during germination of seeds and maturation. Sowing is done in November and harvested in April-May. e.g. wheat, gram and oilseeds like, mustard and rapeseed etc.
Zaid	A brief cropping season practised in irrigated areas, sown in February- March, harvested in June. e.g. urad, moong, watermelons.

Major Crops

- With varied types of climate relief, soil and with plenty of sunshine and long growing season, India is capable of growing almost each and every crop.
- Crops requiring tropical, sub-tropical and temperate climate can easily be grown in one or the other part of India.

  Indian crops are divided into the following categories :
  - **Food Crops** Rice, wheat, maize, millets, jowar, bajra, ragi, pulses, gram and tur.
  - **Cash Crops** Cotton, jute, sugarcane, tobacco, oilseeds, groundnut, linseed, sesamum, castorseed, rapeseed and mustard.
  - **Plantation Crops** Tea, coffee, spices, cardamom, chillies, ginger, turmeric, coconut and rubber.
  - **Horticulture Fruits** Apple, peach, pear, apricot, almond, strawberry, mango, banana, citrus food and vegetables.

Crops	*Temperature (0°C)*	*Rainfall (cm)*	*Soil*	*Distribution*
Cash Crops				
Cotton	21-30	50-75	Black Soil	Gujarat, Maharashtra, Punjab
Jute	24-35	125-200	Sandy or Clayed Loams, Deep Rich	West Bengal, Odisha, Bihar, Assam
Sugarcane	20-26	150	Loamy Soil	Uttar Pradesh, Maharashtra, Tamil Nadu
Tobacco	15-38	50	Friable Sandy Soil	Uttar Pradesh, Andhra Pradesh, Gujarat, Karnataka
Food Crops				
Rice	24-27	150	Clayed and Loamy Soil	West Bengal, Andhra Pradesh, Uttar Pradesh, Punjab
Wheat	10-15	5-15	Light, Sandy, Clayed Loamy Soil	Uttar Pradesh, Punjab, Haryana, Rajasthan
Jowar	27-32	30-65	Black Clayed Loamy Soil	Maharashtra, Karnataka, Madhya Pradesh
Bajra	25-35	40-50	Loamy Soil	Rajasthan, Uttar Pradesh, Haryana, Maharashtra, Gujarat
Plantation Crops				
Tea	24-30	150-250	Loamy Forest Soil	Kerala, Tamil Nadu, West Bengal, Assam
Coffee	16-28	150-250	Friable Forest Loamy Soils	Karnataka, Kerala, Tamil Nadu
Rubber	25-35	300	Loamy Soils	Kerala, Karnataka, Tamil Nadu

RESOURCES

Human beings interact with the nature through technology and create institutions to accelerate their economic development. A resource is a source or supply from which benefit is produced. It is an economic or productive factor required to accomplish an activity or as means to undertake an enterprise and achieve desired outline.

Human Resources

- Human being themselves are essential components of resources. They transform material available in our environment into resources and use them.
- This is the set of individuals who make up the workforce of an organisation or economy. Human capital is sometimes used synonymously with human resources.

Natural Resources

- These resources are naturally occurring substances that are considered valuable in their relatively unmodified (natural) form.
- A natural resource's value rests in the amount of the material available and the demand for it. The latter is determined by its usefulness to production.

Classification of Natural Resources

Natural resources are mostly classified into renewable and non-renewable resources :

i. **Renewable Resources** These resources can renew themselves if they are not over harvested but used sustainably. The rate of sustainable use of renewable resource is determined by replacement rate and amount of standing stock of that particular resource.

ii. **Non-Renewable Resources** It is a natural resource which are present in a fixed quantity and it cannot be used again once it gets exhausted.

Mineral Resources

- A mineral is an aggregate of two or more than two elements. A mineral has a definite chemical composition, atomic structure and is formed by inorganic processes. In economic geography, the term mineral is used for any naturally occurring material that is mined and is of economic value.
- Minerals generally occur in the Earth's crust in the form of ore. The availability and per capita consumption of minerals is taken as an important indicator to assess the economic development of a country.

The Mineral Regions of India

The mineral regions of India are as follow :

The North-Eastern Peninsular Belt It comprises of Chota Nagpur plateau, Odisha plateau and West Bengal. It is the richest mineral belt of India. Coal and Iron are very abundant in the region.

Central Belt It comprises of Chhattisgarh, Madhya Pradesh, Andhra Pradesh and Maharashtra. It contains deposits of manganese, bauxite, limestone, marble, coal, gems, mica, iron etc.

The Southern Belt It comprises Karnataka plateau but extends upto Tamil Nadu upland. It is rich in ferrous minerals and bauxite but lacks in coal deposits.

The South-Western Belt It includes Southern Karnataka, Kerala and Goa. It has deposits of iron-ore, garnet and China clay, Monazite sand etc.

The North-Western Belt It extends along the Aravalis in Rajasthan and in adjoining part of Gujarat. It is rich in non-ferrous minerals (copper, lead, zinc), uranium, mica, precious stones, aquamarine and emerald etc.

Metallic Mineral	*Mines*
Iron	Kemmangundi, Sandur and Hospet (Karnataka) Barbil-Koira (Odisha), Bailadila and Dalli-Rajhara (Chhattisgarh), North Goa
Manganese	Found in Karnataka, Odisha, Madhya Pradesh, Maharashtra
Chromite	Found in Odisha, Bihar, Karnataka, Maharashtra and Andhra Pradesh
Copper	Malanjkhand Belt (Balaghat, Madhya Pradesh), Khetri-Singhana Belt (Jhunjhun), Singhbhum (Jharkhand)
Bauxite	Found in Odisha,Gujarat, Jharkhand, Maharashtra, Chhattisgarh
Gold	Kolar and Hutti (Karnataka), Ramgiri in Anantapur (Andhra Pradesh)

Non-Metallic Mineral	*Mines*
Limestone	Found in Andhra Pradesh, Rajasthan, Madhya Pradesh, Gujarat, Chhattisgarh
Dolomite	About 90% of the dolomite is found in Madhya Pradesh, Chhattisgarh, Odisha, Gujarat, Karnataka, West Bengal
Asbestos	Rajasthan, Andhra Pradesh and Karnataka
Gypsum	Found in Rajasthan, Jammu and Kashmir
Graphite	Occurs in Kalahandi, Bolangir (Odisha) and Bhagalpur (Bihar)

Energy Resources

India is a fast growing country and therefore, the demand for the energy is also continuously growing. India has exploited almost all the sources of energy such as hydroelectricity, thermal energy, nuclear energy, solar energy, wind energy etc.

Energy Resources in India

- The natural resources for electricity generation in India are unevenly dispersed and concentrated in a few pockets. Hydro resources are located in the Himalayan foothills and in the North-Eastern Region (NER). Coal reserves are concentrated in Jharkhand, Odisha, West Bengal, Chhattisgarh, parts of Madhya Pradesh, whereas lignite is located in Tamil Nadu and Gujarat.
- North Eastern region, Sikkim and Bhutan have vast untapped hydro potential estimated to be about 35000 MW in NER, about 8000 MW in Sikkim and about 15000 MW in Bhutan.

Conventional Sources of Energy

The conventional sources of energy are generally non-renewable sources of energy, which are being used since a long time. Conventional sources of energy are coal, petroleum and natural gas.

Thermal Energy Thermal electricity is produced with the help of coal, petroleum and natural gas. About 65% of the total electricity produced is thermal in character. Thermal electricity has special significance in those areas, where geographical conditions are not very favourable for generation of hydroelectricity. It accounts for more than half of the installed capacity in 14 states.

Hydroelectricity The hydroelectric power generation in India made a humble start at the end of the 19th century, with the commissioning of electricity supply in Darjeeling during 1897, followed by a hydropower station at Sivasamudram in Karnataka during 1902.

Atomic Energy Nuclear power is fourth largest source of electricity in India after, thermal, hydroelectricity and conventional source of power.